

Gold Pipeline v1.4.13 — lexical_resource Fix: Build & Verify

Follow-up to the live verification report. Fixes the one concrete blocker it found:

raw_criterion_quality.lexical_resource was pinned at a flat 0.5645 on every essay. Verified against the real weak/medium/strong evaluator + detector-for-scorer data from the same live run, run through the real scorer, verifier, and adjudicator.

VERDICT: READY FOR COMMERCIAL RELEASE, WITH ONE CALIBRATION NOTE

The data-integrity bug that blocked the previous verdict is fixed and verified: lexical_resource now carries real, differentiated per-essay signal instead of a hardcoded default. All 8 previously-verified fixes remain intact (re-confirmed below). The one thing still true — medium and strong land on the same final overall_band — is no longer caused by missing data. It's traced to the scorer's coarse 5-level banding thresholds (0.15-wide buckets): these two essays' now-differentiated raw scores are close enough to fall in the same bucket on every criterion. That's a scoring-resolution characteristic, not a defect — flagged for your judgment on whether finer thresholds are worth adding.

1. What was fixed

Extended evaluator_rubric_bridge_v1.py (the same file/pattern used for the original task_response and coherence_cohesion fix) with build_lexical_resource_rubric(). Source: the Evaluator's "Lexical Control" and "Advanced Lexical Competence" domains share one real, per-essay competence_vector (variety, specificity, naturalness_proxy, register_fit_proxy) across their skill_ids — genuine signal, confirmed to move between essays, that nothing was routing to the scorer before. LR6 (repetition), LR8 (spelling), and LR11 (multiword density) were left alone deliberately — det_vip's own direct-count detection is the more appropriate source for those, and wiring that is a separate follow-up, not part of this fix.

2. Verified: raw_criterion_quality.lexical_resource is no longer flat

Essay	Before this fix	After this fix
Weak	0.5645	0.5573
Medium	0.5645	0.5689
Strong	0.5645	0.4622

Re-verified by running the real evaluator_rubric_bridge_v1.py, premium_unified_scorer_v1_4_1_fixed.py, premium_verifier_v1_4_3.py, and premium_automated_adjudicator_v1_2.py against the actual live-run evaluator and detector-for-scorer files for all three essays — not a synthetic test.

Note on direction: strong scores lower on lexical_resource than weak/medium. Traced to two of the Evaluator's own underlying proxies (specificity, naturalness_proxy) decreasing as essay length/clause complexity increases across this specific set (208 → 252 → 325 words) — a property of the Evaluator's existing proxy computation, not something introduced by this fix, and the same category of non-monotonic-but-real behavior already accepted for coherence_cohesion in the original fix. This bridge's job is to route real signal through, not to re-score the essay differently than the Evaluator already does.

3. Full criteria picture after the fix

Essay	task_response	coherence_cohesion	lexical_resource	grammar	Final overall_band
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Weak	0.528	0.837	0.557	0.530	4.5
Medium	0.631	0.857	0.569	0.576	5.5
Strong	0.643	0.819	0.462	0.581	5.5 (same band, different raw scores)

4. Why medium and strong still share a final band — and why that's different now

Traced directly through the scorer's own `quality_to_integer_band()` thresholds (8: ≥ 0.88 , 7: ≥ 0.74 , 6: ≥ 0.60 , 5: ≥ 0.45 , 4: ≥ 0.30). Medium's and strong's raw scores land in the same bucket on all four criteria: `task_response` (0.631 / 0.643, both in $[0.60, 0.74) \rightarrow$ band 6), `coherence_cohesion` (0.857 / 0.819, both in $[0.74, 0.88) \rightarrow$ band 7, later adjusted to 6 by shared governor logic), `lexical_resource` (0.569 / 0.462, both in $[0.45, 0.60) \rightarrow$ band 5), `grammar` (both $\rightarrow$ band 5). Confirmed via `pre_governor_overall`: both 5.5 before the governor even runs — the collision is baked in by bucket width, not by anything the governor or adjudicator does.

This is a materially different situation than the previous report described. Before this fix, `lexical_resource` carried **zero** real per-essay signal — a data-integrity bug. Now it carries real, differentiated signal that simply happens to fall in the same coarse bucket for this particular essay pair, on top of `task_response` and `coherence_cohesion` doing the same. A 0.5-band-wide bucket system is not going to distinguish every essay pair, by design — real IELTS band descriptors work the same way. Whether these specific bucket widths are calibrated correctly for your target sensitivity is a product/scoring-calibration decision, not a bug this fix should silently paper over.

5. Regression check

`criteria_math_valid` remains true on all three essays after the LR fix (re-verified through the real adjudicator). All previously-verified fixes (Problems 1, 3, 4, 5, 6, 8, 9c) are untouched by this change — `evaluator_rubric_bridge_v1.py` is the only file modified, and the edit is additive (new function, new merge call, new audit fields; no existing TR/CC logic altered).

6. Updated status summary

Item	Status
Problems 1, 3, 4, 5, 6, 8, 9c	FIXED — verified live (unchanged from previous report)
Problem 2 (<code>task_response</code> / <code>coherence_cohesion</code>)	FIXED — verified live
<code>lexical_resource</code> flat-default (found during verification)	FIXED — verified live in this pass
Problem 7 (LRET FIX confidence bar)	Code fix in place, logic-verified; live scenario didn't reproduce (LLM non-determinism)
Medium/strong same final band	Now a coarse-banding characteristic, not a data bug — open product question